

HR WORKFORCE ANALYTICS PROJECT

Employee Attrition & HR Analytics Dashboard

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Excel

SQL

Python

Power BI

Project Overview



Business Problem

Losing employees is expensive — recruiting, onboarding, and lost productivity all add cost. But a single company-wide attrition number doesn't tell HR where to act.



Project Goal

Build an end-to-end HR analytics solution — from raw data to an interactive dashboard — that identifies attrition drivers and enables targeted, data-driven retention decisions.

WHY HR ANALYTICS MATTERS

Replacing an employee can cost 6–9 months of their salary once recruiting, onboarding, and lost productivity are factored in. HR analytics turns scattered employee records into a decision-making tool — helping organizations move from reacting to attrition after it happens, to predicting and preventing it before it does.

Project Objectives

01

Analyze Employee Attrition

Break down a flat company-wide attrition rate into department, tenure, age, and experience-level detail.

02

Identify High-Risk Employee Groups

Pinpoint exactly which segments of the workforce carry the highest attrition risk.

03

Evaluate Workforce Performance

Examine performance ratings, compensation, and training investment across departments.

04

Support Data-Driven HR Decisions

Translate findings into a dashboard and recommendations HR can act on directly.

Dataset Overview

DATASET NAME

IBM HR Analytics
Employee Attrition & Performance

4,327 Employees

31 Features / Attributes

Kaggle Data Source

Source: IBM HR Analytics Employee Attrition & Performance dataset (Kaggle)

KEY COLUMNS

EmployeeID	Unique employee identifier
Attrition	Target variable — Yes / No
Department	Sales, R&D, HR
Age	Employee age
Gender	Male / Female
JobRole	10 distinct roles
MonthlyIncome	Compensation
WorkLifeBalance	1-4 rating scale
JobSatisfaction	1-4 rating scale
TotalWorkingYears	Career tenure
PerformanceRating	1-5 rating scale
MaritalStatus	Single / Married / Divorced

HOW IT WAS BUILT

End-to-End Workflow



SQL Analysis

WHAT WAS ANALYZED

● Employee Workforce Analysis

● Attrition Analysis

● Salary & Performance Analysis

● Department Insights

● Advanced SQL Analytics

TECHNIQUES USED

30 Business Queries

Window Functions

Common Table Expressions (CTE)

CASE Statements

Aggregate Functions

From simple aggregations to window functions (RANK, DENSE_RANK, ROW_NUMBER), CTEs, and a composite CASE-based HR risk score — the SQL layer built the analytical foundation the dashboard was designed on top of.

Python Analysis



Data Cleaning

Removed duplicates, standardized inconsistent category labels, and fixed data-type mismatches.



Data Preprocessing

Bucketed continuous fields (age, tenure) and mapped numeric scales to readable labels.



Missing Value Handling

Identified and treated null/missing entries so downstream analysis wasn't skewed.



Exploratory Data Analysis (EDA)

Explored distributions and relationships across attrition, department, and satisfaction.



Statistical Analysis

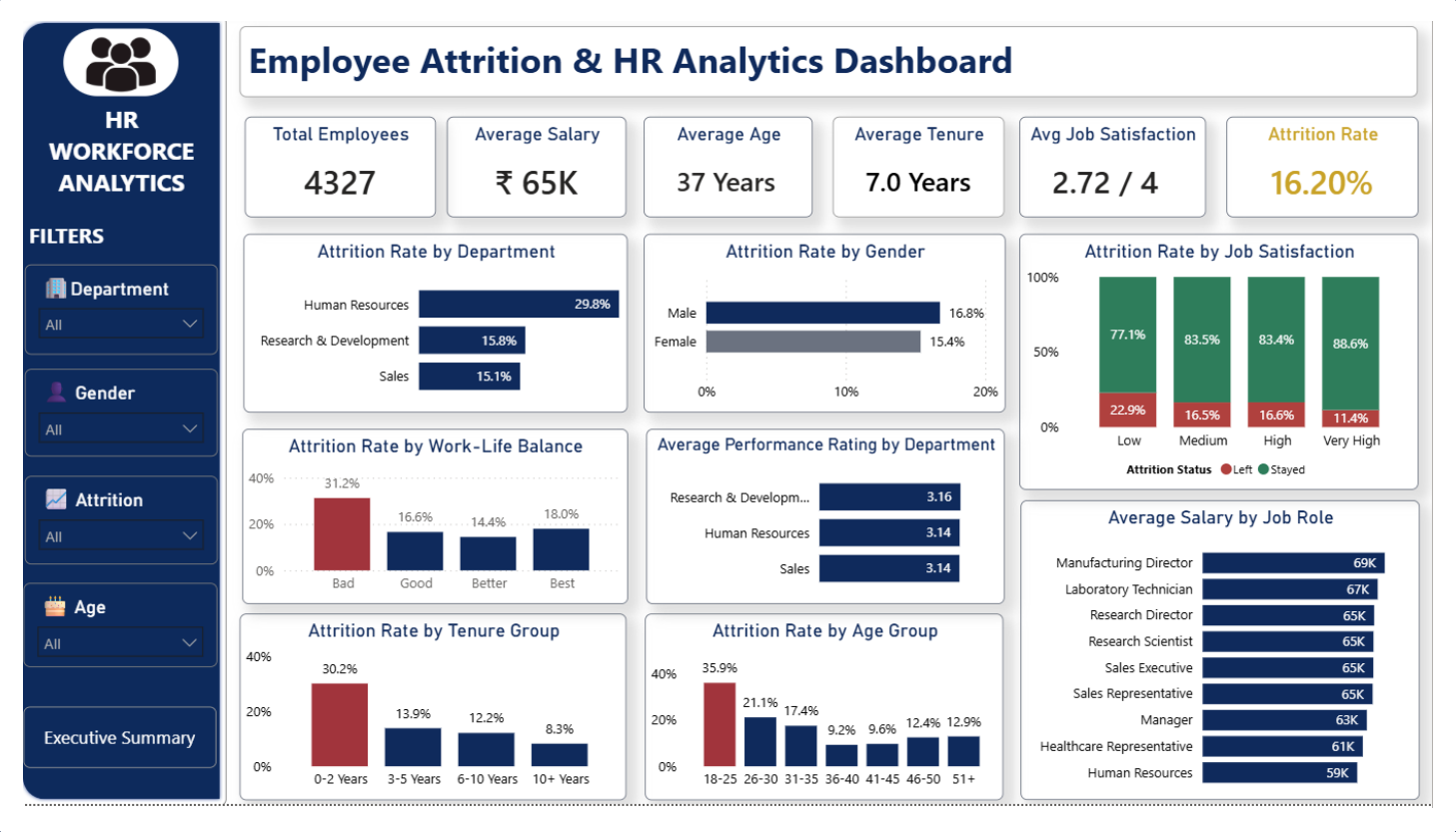
Quantified correlations between attrition and tenure, age, satisfaction, and work-life balance.



Data Visualization

Built exploratory charts to validate patterns before they were built into the dashboard.

Dashboard Overview



KPI Cards

Six headline metrics — headcount, salary, age, tenure, satisfaction, and attrition rate.

Filters

Department, Gender, Education, Attrition, Job Role, Marital Status, and Age slicers.

Charts

Attrition broken down by department, gender, satisfaction, work-life balance, and tenure.

Navigation

A dedicated button links to the Executive Summary page for stakeholder review.

Executive Summary

A condensed, decision-oriented second page for non-technical stakeholders.

Key Business Insights

16.2%	Overall Attrition Rate The company-wide baseline every other finding compares against.
29.8%	HR Has the Highest Attrition Almost double Research & Development (15.8%) and Sales (15.1%).
30.2%	0–2 Year Tenure = Highest Risk Nearly 4× the attrition rate of employees with 10+ years (8.3%).
35.9%	Younger Employees Attrite More Ages 18–25 attrite at more than double the 26–30 cohort (21.1%).
31.2%	Poor Work-Life Balance Drives Attrition Vs. 16.6% for employees reporting “Good” work-life balance.

Business Recommendations



Improve Onboarding Programs

Structured onboarding and mentoring during an employee's first two years, when attrition risk is highest.



Enhance Work-Life Balance

Flexible work policies and wellness initiatives to address the strongest experience-related driver.



Increase Employee Engagement

Regular feedback, career development conversations, and recognition — especially for early-career staff.



Review Retention Strategies

A focused retention review within Human Resources, given its disproportionate attrition rate.



Monitor High-Risk Employees

Use the HR Risk Score to proactively flag and support employees before they leave.

Skills Demonstrated

TECHNICAL SKILLS

- Excel
- SQL
- Python
- Power BI
- DAX
- Power Query

ANALYTICAL SKILLS

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Dashboard Development
- KPI Reporting
- Business Intelligence

WRAPPING UP

Conclusion

- **Project Summary**

An end-to-end HR analytics solution — Excel to SQL to Python to Power BI — that turns a flat 16.2% attrition rate into a segment-level view HR can act on.

- **Business Impact**

Identifies exactly where retention efforts should focus first: early-tenure employees, younger staff, and the HR department itself.

- **Key Learnings**

End-to-end pipeline thinking, SQL window functions and CTEs, DAX measure debugging, and translating analysis into stakeholder-ready visuals.

Thank You